# GIWON SONG

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## EDUCATION

**Seoul National University**

Bachelor of Science in Electronic and Computer Engineering — 2029 (expected)

- Cumulative GPA: 4.04 / 4.3
- Academic Excellence Scholarship (Merit-based) : 2023 – 2024

**University of California, Berkeley**

Visiting International Exchange Student (Physics Major) — Jan 2027 – May 2027

**Korea Science Academy of KAIST (KSA)**

High School Diploma — 2023

## RESEARCH EXPERIENCE

**Republic of Korea Army (ROKA)**

Cryptanalyst — May 2025 – Nov 2026

- Decrypted complex ciphers by training a large language model (LLM) targeted to specific properties of the plaintexts.

**Quantum Information and Quantum Computer Lab (QuIQCL), Seoul National University**

Undergraduate Research Intern — July 2024 – Apr 2025

- Conducted research on the theoretical limits of quantum computing and algorithmic efficiency.

Undergraduate Research Opportunity Program (UROP) Participant — July 2023 – Aug 2023

- Co-developed IQUIP (Ion trap Quantum computing User Interface Project). GitHub : [GitHub]

**Korea Science Academy of KAIST (KSA)**

Student Researcher — 2021

- Researched advanced implementations of Neural Radiance Fields (NeRF) for multiple objects. GitHub : [GitHub]

## PUBLICATIONS

**Published / Pre-prints**

- Hwang, Y.*, **Song, G.***, Park, K., Kwon, H.†, Kim, T.† "*Efficient expectation value estimation for quantum circuits via extended stabilizer frameworks and adaptive variance estimation.*" ArXiv preprint, 2026.
  | ArXiv : [ArXiv] | GitHub : [GitHub]
- Park, K., Choi, K., Kim, M., **Song, G.**, Kim, T.† "*Quantum Linear Multistep Method for Using a Quantum Oracle with Differential Equations.*" ArXiv preprint, 2025.
  | ArXiv : [ArXiv]

**These authors contributed equally to the work. †Corresponding author(s).*

## AWARDS & HONORS

- **Special Prize (IBM Quantum)** | Korea Q Hackathon — 2024
- **Advanced Badge (Completed all exercises)** | IBM Quantum Spring Challenge — 2024
- **Director's Award, Quantum Information Research Support Center** | Korea Q Hackathon — 2023
- **Silver Prize** | Korea Olympiad in Informatics (KOI) — 2016

## SKILLS & LANGUAGES

**Technical skills**: C++, Python, Qiskit, LaTeX, Git

**Languages**: Korean (native), English (professional working proficiency), Japanese (professional working proficiency)

- iBT TOEFL: 5.5/6 (110/120)
- JLPT: N1 (127/180), N2 (165/180)